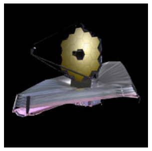

7648 - N6946-BH1: Fading following a failed supernova, or post-merger recovery?

Cycle: 4, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
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Dr. Ben Davies (CoI) (ESA Member)	Liverpool John Moores University

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
MIRI				
	2	MIRI	MIRI Imaging	(1) NAME-N6946-BH1
	3	NIRCam	NIRCam Imaging	(1) NAME-N6946-BH1
Observation Folder				
	4	MIRI MRS	MIRI Medium Resolution Spectroscopy	(1) NAME-N6946-BH1

ABSTRACT

Recent JWST imaging revealed a luminous infrared source at the location of failed supernova (SN) candidate N6946-BH1. While significantly fainter than the progenitor (13-25% of the bolometric luminosity), the spectral energy distribution (SED) is consistent with both a failed SN and a stellar merger or outburst event. Curiously, observations taken one month later in matching filters revealed a 50% flux reduction in the shortest MIRI filter (5.6m). The mid-IR fluxes also appear to resemble Case C Polycyclic Aromatic Hydrocarbon (PAH) emission, typically produced when dust is irradiated by a UV field.

We are requesting a total of 16 hours of JWST time in Cycle 4: 5 hours for continued monitoring of N6946-BH1 to assess variability across the near- and mid-IR, and 11 hours for medium-resolution MIRI spectroscopy to search for PAH features. In addition, we are requesting another 5 hours in Cycle 5 to repeat the photometry observations so that we can determine the rate of change in brightness of the source. If the near-IR flux continues to fade, it will strongly suggest the formation of a black hole. If the source remains constant or brightens in the near-IR while fading in the mid-IR, this will support a merger scenario, with the dust clearing to reveal the surviving star. These observations will not only constrain the timescales for such events but also help estimate the ejected dust mass from mid-IR emission, ultimately resolving a long-standing debate over the fate of N6946-BH1.

OBSERVING DESCRIPTION

We are proposing to image the remaining infrared source remaining at the position of failed SN candidate N6946-BH1 with NIRCam and MIRI over two epochs. This will enable us to understand how the source changes in brightness across the SED with time, and will allow us to constrain the total dust mass of the system. Ultimately these observations will constrain the fate of BH1: has it formed a black hole, or is there a surviving star following a merger?

We are also requesting 10 hours of JWST time to obtain a MIRI MRS spectrum of the remaining source at the position of failed SN candidate BH1. The MRS spectrum will reveal whether or not there are PAH features present in the SED, and break the degeneracy between the merger and failed SN scenarios.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NAME-N6946-BH1	RA: 20 35 27.5600 (308.8648333d) Dec: +60 08 8.29 (60.13564d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Black holes, Circumstellar dust, M supergiants, Red supergiants]
Extended=NO

Observation

Proposal 7648, Observation 2: MIRI

Wed Jun 17 16:00:17 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NAME-N6946-BH1	RA: 20 35 27.5600 (308.8648333d) Dec: +60 08 8.29 (60.13564d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Black holes, Circumstellar dust, M supergiants, Red supergiants]
Extended=NO

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	CYCLING	1	4		1	1			DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F560W	SLOWR1	21	1	1	Dither 1	4	4	2006.753	
2	F770W	SLOWR1	11	1	1	Dither 1	4	4	1051.156	
3	F1000W	SLOWR1	11	1	1	Dither 1	4	4	1051.156	
4	F1280W	SLOWR1	11	1	1	Dither 1	4	4	1051.156	
5	F1800W	SLOWR1	11	1	1	Dither 1	4	4	1051.156	
6	F2100W	SLOWR1	11	1	1	Dither 1	4	4	1051.156	
7	F2550W	SLOWR1	11	1	1	Dither 1	4	4	1051.156	

Observation

Proposal 7648, Observation 3: NIRCam

Wed Jun 17 16:00:17 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NAME-N6946-BH1	RA: 20 35 27.5600 (308.8648333d) Dec: +60 08 8.29 (60.13564d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Black holes, Circumstellar dust, M supergiants, Red supergiants]
Extended=NO

Template

Module	Subarray
B	FULL

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRASCA	4	STANDARD	8" (SMALL)	1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F115W	F250M	MEDIUM8	10	1	4	4	4208.814	
2	F182M	F360M	SHALLOW4	10	1	4	4	2104.407	

Special Requirements

Aperture PA Range 30 to 260 Degrees (V3 29.94737309 to 259.94737309)
Offset -14.0 arcsec, -14.0 arcsec

Observation

Proposal 7648, Observation 4: MIRI MRS

Wed Jun 17 16:00:17 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(MIRI MRS (Obs 4)) Warning (Form): Imager Filter overlap.

(Visit 4:1) Warning (Form): Data Excess over middle threshold

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NAME-N6946-BH1	RA: 20 35 27.5600 (308.8648333d) Dec: +60 08 8.29 (60.13564d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Black holes, Circumstellar dust, M supergiants, Red supergiants]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	F1000W	FAST	8	1	1	22.2	169847.1

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F560W	FASTR1	200	1	1	Dither 1	4	4	2220.032	
1	SHORT(A)	MRSLONG		FASTR1	200	4	1	Dither 1	4	16	8913.428	
1	SHORT(A)	MRSSHORT		FASTR1	200	4	1	Dither 1	4	16	8913.428	
2		IMAGER	F1000W	FASTR1	200	1	1	Dither 1	4	4	2220.032	
2	MEDIUM(B)	MRSLONG		FASTR1	200	4	1	Dither 1	4	16	8913.428	
2	MEDIUM(B)	MRSSHORT		FASTR1	200	4	1	Dither 1	4	16	8913.428	
3		IMAGER	F2550W	FASTR1	200	1	1	Dither 1	4	4	2220.032	
3	LONG(C)	MRSLONG		FASTR1	200	3	1	Dither 1	4	12	6682.296	
3	LONG(C)	MRSSHORT		FASTR1	200	3	1	Dither 1	4	12	6682.296	